TEIJIN CARBON

Product overview

Tenax™ and **Tenax Next™ filament yarn**

Fiber family		Sizing properties	Filament count	Yield (without sizing)	Additional information	Tensile strength		Tensile modulus internal test method		Tensile modulus ASTM / SACMA		Elongation at break	Sizing	Size level	Density		Filament diameter
						[MPa]	[ksi]	[GPa]	[Msi]	[GPa]	[Msi]	[%]		[%]	[g/cm³]	[lb/in³]	[µm]
HTA	**HTA40**	**E15**	**1K**	**67tex**	**15S**	4100	590	240	35	230	33	1.7	EP	2.5	1.77	0.064	7.0
	HTA40	**E13**	**3K**	**200tex**									EP	1.3			
	HTA40	**E13**	**3K**	**200tex**	**15Z**								EP	1.3			
	HTA40	**E13**	**6K**	**400tex**									EP	1.3			
	HTA40	**E13**	**6K**	**400tex**	**10Z**								EP	1.3			
	HTA40	**E13**	**12K**	**800tex**									EP	1.3			
HTS	**HTS40**	**E13**	**3K**	**200tex**		4400	640	240	35	230	33	1.8	EP	1.3	1.77	0.064	7.0
	HTS40	**E13**	**6K**	**400tex**									EP	1.3			
	HTS40	**F13**	**12K**	**800tex**									PU	1.0			
	HTS40	**F13**	**12K**	**800tex**	**10Z**								PU	1.0			
	HTS40	**F13**	**24K**	**1600tex**									PU	1.0			
	HTS45	**E23**	**3K**	**200tex**		4500	650	245	36	240	35	1.8	EP	1.2	1.76	0.064	7.0
	HTS45	**E23**	**3K**	**200tex**	**15Z**								EP	1.2			
	HTS45	**E23**	**12K**	**800tex**				240	35	235	34	1.9	EP	1.3	1.77	0.064	
	HTS45	**E23**	**12K**	**800tex**	**10Z**								EP	1.3			
	HTS45	**E23**	**24K**	**1600tex**		4800	700	240	35	235	34	2.0	EP	1.2	1.79	0.065	
	HTS45*	**E23**	**24K**	**1600tex**									EP	1.2			
STS	**STS40**	**E23**	**24K**	**1600tex**		4300	620	250	36	245	36	1.7	EP	1.3	1.78	0.064	7.0
	STS40	**F11**	**24K**	**1600tex**				240	35	235	34	1.8	PU	0.17			
	STS40	**F13**	**24K**	**1600tex**				240	35	235	34	1.8	PU	1.0			
	STS40	**F13**	**24K**	**1600tex**	**S**								PU	1.0			
	STS40	**F13**	**48K**	**3200tex**				250	36	245	36	1.7	PU	1.0	1.77		
	STS40	**E23**	**48K**	**3200tex**									EP	1.0	1.78		
UTS	**UTS50**	**C31**	**24K**	**1600tex**		5100	740	245	36	235	34	2.1	PA	0.4	1.81	0.065	7.0
	UTS50	**F13**	**12K**	**800tex**									PU	1.0	1.78	0.064	
	UTS50	**F22**	**12K**	**800tex**	**S**								PU	0.9			
	UTS50	**F24**	**24K**	**1600tex**	**D**			245/250					PU	1.6/2.1	1.81	0.065	
ITS	**ITS50**	**F23**	**24K**	**1600tex**	**D**	5100	740	265	38	255	37	1.9	PU	1.0	1.80	0.065	7.0
	ITS55	**E23**	**24K**	**1600tex**				280	41	270	39	1.8	EP	1.2	1.75	0.063	
IMS	**IMS60**	**E13**	**24K**	**830tex**		5800	840	290	42	280	41	2.0	EP	1.3	1.79	0.065	5.0
	IMS65	**E23**	**24K**	**830tex**		6000	870					2.1	EP	1.3	1.78	0.064	
	IMS65	**E23**	**36K**	**1630tex**		5800	840	280	41	270	39	2.1	EP	1.3	1.80	0.065	5.7
	IMS65	**E23**	**36K**	**1630tex**	**S**								EP	1.3			
	IMS6L	**E23**	**24K**	**830tex**		5900	855	280	41	270	39	2.1	EP	1.3	1.80	0.065	5.0
UMS	**UMS40**	**F23**	**24K**	**800tex**	**S**	4700	680	390	57	375	54	1.2	PU	1.0	1.79	0.065	4.9
	UMS45	**F22**	**12K**	**385tex**		4600	670	425	62	415	60	1.1	PU	0.9	1.83	0.066	4.7

*Tenax Next™

Notes

- Variations in test results may occur due to origin-specific testing procedures.
- Abbreviations:
  - **D** fixed wide band
  - **S** soft wide band
  - **xS/Z** twist counts per meter and twist direction

Reinforcing excellence

Teijin Limited
customer@teijin.co.jp

Teijin Carbon Europe GmbH
sales@teijincarbon.com

Teijin Carbon America, Inc.
info@teijincarbon.us

www.teijincarbon.com

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Product overview

Tenax™ and Tenax Next™ filament yarn

Tube properties

		Japan		Germany	USA
Tube length	[mm]	182	280	290	280
Tube internal diameter	[mm]	77	77	77	77
Tube weight	[g]	110	160	190	175
Stroke	[mm]	152	254	254	254

Bobbin properties

Net weight	[kg]	0.5	1.0	2.0	4.0	6.0	8.0	10.0
External bobbin diameter	[mm]	105	120	145	180	205	225	245

Bobbins per packaging

	[kg]	0.5	1.0	2.0	4.0	6.0	8.0	10.0
Carton	Japan	20/24	20	12	6	6	-	4
	Germany	-	16	11/12	6	6	3	-
	USA	-	-	12	8	5	4	-
Large packaging	Japan	-	-	-	-	75	60	54
	Germany	-	-	150	90	69	60	-
	USA	-	-	-	-	69	60	-

Packaging (L x W x H)

Carton	Japan		[mm]	640 x 510 x 300
		(4 kg)	[mm]	550 x 370 x 300
		(6 kg - 10 kg)	[mm]	570 x 570 x 320
	Germany		[mm]	600 x 420 x 320
	USA		[mm]	610 x 510 x 300
Large packaging incl. pallet	Japan		[mm]	1200 x 1100 x 1050
	Germany		[mm]	1000 x 1200 x 910
	USA		[mm]	1020 x 1220 x 1150

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customer@teijin.co.jp

Teijin Carbon Europe GmbH
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